

Mock Midterm 2 — A

Worked solutions and marking guide

18 points: Exercise 1 (10), Exercise 2 (8)

These are full solutions, not only final answers. The marking allocations are for self-assessment, not an official course rubric. Equivalent derivations and the opposite sign of a PCA eigenvector are valid. Give partial credit for a correct method even if a later arithmetic error propagates.

Exercise 1: Two sensors, one class-dependent rate

1a Joint likelihood and constrained estimation

[3 p]

First use the product rule $p(x, C_k) = \pi_k p(x | C_k)$, then conditional independence of the two sensors. Independence of training observations gives

$$p(X, T | \pi, \lambda) = \prod_{n=1}^N \prod_{k=1}^K \left[\pi_k \frac{2^{m_n}}{m_n!} \lambda_k^{m_n+1} e^{-\lambda_k(2+s_n)} \right]^{t_{nk}}.$$

In particular, the two rate factors combine to $\lambda_k^{m_n+1}$. Writing $B = \sum_n [m_n \log 2 - \log(m_n!)]$, which is parameter-independent,

$$\ell = B + \sum_k [N_k \log \pi_k + (M_k + N_k) \log \lambda_k - (2N_k + S_k) \lambda_k].$$

For the rates,

$$\frac{\partial \ell}{\partial \lambda_k} = \frac{M_k + N_k}{\lambda_k} - (2N_k + S_k) = 0 \implies \boxed{\hat{\lambda}_k = \frac{M_k + N_k}{2N_k + S_k}}.$$

The second derivative is $-(M_k + N_k)/\lambda_k^2 < 0$. The log-likelihood tends to $-\infty$ at both ends of the positive rate domain, so this is its unique global rate maximum. The positive count N_k ensures both numerator and denominator are positive, even when every count or waiting time in a class is zero.

For the priors use

$$\mathcal{L} = \sum_k N_k \log \pi_k + \eta \left(\sum_k \pi_k - 1 \right).$$

The stationary conditions give $N_k/\pi_k + \eta = 0$, hence $\pi_k = -N_k/\eta$. Summing gives $\eta = -N$ and therefore

$$\boxed{\hat{\pi}_k = N_k/N}.$$

The prior part is strictly concave on the simplex interior and tends to $-\infty$ if any $\pi_k \rightarrow 0$ with $N_k > 0$.

Suggested credit: 1 p for the likelihood and log-likelihood; 1 p for the rate estimate and maximum check; 1 p for the constrained prior estimate.

1b Posterior scores and class region

[2 p]

The fitted parameters are

$$\boxed{(\hat{\pi}_1, \hat{\pi}_2, \hat{\pi}_3) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4}), \quad (\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3) = (1, 2, \frac{1}{2})}.$$

Bayes' rule gives

$$p(C_k | m, s) = \frac{\pi_k \lambda_k^{m+1} e^{-\lambda_k(2+s)}}{\sum_j \pi_j \lambda_j^{m+1} e^{-\lambda_j(2+s)}}.$$

The common factor $2^m/m!$ cancels. An affine score is

$$g_k(m, s) = \log \pi_k + \log \lambda_k - 2\lambda_k + m \log \lambda_k - s\lambda_k, \quad p(C_k | m, s) = \text{softmax}(g)_k.$$

Thus C_1 is at least as probable as its two competitors precisely when

$$\begin{aligned} g_1 - g_2 &= 2 + s - m \log 2 \geq 0, \\ g_1 - g_3 &= (m + 2) \log 2 - 1 - \frac{1}{2}s \geq 0. \end{aligned}$$

Equivalently,

$$\boxed{m \log 2 \leq 2 + s, \quad 1 + \frac{1}{2}s \leq (m + 2) \log 2}.$$

With $m, s \geq 0$ treated as real coordinates, this region is an intersection of half-spaces and is convex. The actual count-domain subset is not a convex subset of $\mathbb{R}^2$ merely because it is restricted to integer m ; the question explicitly asks about the continuous relaxation. Ties may be assigned by any specified tie rule, whereas these non-strict inequalities describe posterior dominance.

Suggested credit: 0.5 p for parameters; 0.5 p for scores/posterior; 0.75 p for inequalities; 0.25 p for convexity with the domain qualification.

1c A missing sensor changes the decision

[1 p]

For the complete input $(0, \frac{1}{2})$,

$$g_1 - g_2 = \frac{5}{2} > 0, \quad g_1 - g_3 = 2 \log 2 - \frac{5}{4} > 0.$$

Consequently the predicted class is $\boxed{C_1}$.

If the time is unobserved, integrate out the time density:

$$p(m | C_k) = \int_0^\infty p(m, s | C_k) ds = \frac{(2\lambda_k)^m e^{-2\lambda_k}}{m!},$$

since the exponential density integrates to one. Missingness itself gives no additional class evidence under the stipulated independent failure mechanism. The count-only posterior is

$$\boxed{p(C_k | m) = \frac{\pi_k \lambda_k^m e^{-2\lambda_k}}{\sum_j \pi_j \lambda_j^m e^{-2\lambda_j}}}.$$

At $m = 0$ the unnormalized probabilities are

$$\frac{1}{2}e^{-2}, \quad \frac{1}{4}e^{-4}, \quad \frac{1}{4}e^{-1}.$$

The last is largest: its ratio to the first is $e/2 > 1$. The new prediction is $\boxed{C_3}$. Setting $s = 0$ retains the factor $p(s = 0 | C_k) = \lambda_k$ as a density value and gives the complete-input prediction C_1 , not the count-only prediction. A missing observation is not an observed zero.

Suggested credit: 0.5 p for marginalization and the count-only posterior; 0.5 p for both decisions and the zero-imputation distinction.

1d Conditional independence is not marginal independence**[1 p]**

Within a class,

$$\mathbb{E}[MS \mid C_k] = \mathbb{E}[M \mid C_k]\mathbb{E}[S \mid C_k] = 2.$$

Thus $\mathbb{E}[MS] = 2$, while

$$\mathbb{E}[M] = 2 \sum_k \pi_k \lambda_k, \quad \mathbb{E}[S] = \sum_k \frac{\pi_k}{\lambda_k}.$$

In general this gives

$$\text{Cov}(M, S) = 2 - 2 \left(\sum_k \pi_k \lambda_k \right) \left(\sum_k \frac{\pi_k}{\lambda_k} \right).$$

For these fitted parameters,

$$\sum_k \pi_k \lambda_k = \frac{9}{8}, \quad \sum_k \frac{\pi_k}{\lambda_k} = \frac{9}{8}, \quad \boxed{\text{Cov}(M, S) = 2 - \frac{81}{32} = -\frac{17}{32}}.$$

Nonzero covariance rules out unconditional independence. Classes with higher rates tend to have higher counts but shorter waiting times; the hidden class links the sensors after it is marginalized out.

Suggested credit: 0.5 p for the expectation/covariance derivation; 0.5 p for the value and the independence conclusion.

1e MAP with correlated weight regularization**[2 p]**

The label likelihood is $\prod_{n,k} y_{nk}^{t_{nk}}$. Bayes' rule gives

$$p(W \mid \Phi, T) \propto \left(\prod_{n,k} y_{nk}^{t_{nk}} \right) \exp \left[-\frac{\alpha}{2} \sum_k w_k^\top R w_k \right].$$

The evidence and Gaussian normalizers do not depend on W . Minimize

$$\boxed{L(W) = - \sum_{n,k} t_{nk} \log y_{nk} + \frac{\alpha}{2} \sum_k w_k^\top R w_k}.$$

Let $a_{nk} = \sum_r w_{kr} \phi_{nr}$. In index notation,

$$\frac{\partial \log y_{nk}}{\partial w_{jr}} = \frac{\partial}{\partial w_{jr}} \left(a_{nk} - \log \sum_\ell e^{a_{n\ell}} \right) = (\delta_{kj} - y_{nj}) \phi_{nr}.$$

Since each target row sums to one, the likelihood contribution is

$$- \sum_{n,k} t_{nk} (\delta_{kj} - y_{nj}) \phi_{nr} = \sum_n (y_{nj} - t_{nj}) \phi_{nr}.$$

For the prior contribution, symmetry of R is essential:

$$\begin{aligned} \frac{\partial}{\partial w_{jr}} \frac{\alpha}{2} \sum_{p,q} w_{jp} R_{pq} w_{jq} &= \frac{\alpha}{2} \left(\sum_q R_{rq} w_{jq} + \sum_p w_{jp} R_{pr} \right) \\ &= \alpha \sum_q R_{rq} w_{jq}. \end{aligned}$$

Therefore

$$\boxed{\frac{\partial L}{\partial w_{jr}} = \sum_n (y_{nj} - t_{nj}) \phi_{nr} + \alpha \sum_q R_{rq} w_{jq}},$$

$$\nabla_{w_j} L = \sum_n (y_{nj} - t_{nj}) \phi_n + \alpha R w_j \in \mathbb{R}^3.$$

The numerator-layout derivative is the transpose, a 1×3 row.

Suggested credit: 0.5 p for the objective; 1 p for the component calculation including the full R ; 0.5 p for the assembled gradient.

1f Softmax ambiguity and the MAP representative

[1 p]

For a common shift v , each logit gains $v^\top \phi_n$:

$$\frac{e^{(w_k+v)^\top \phi_n}}{\sum_j e^{(w_j+v)^\top \phi_n}} = \frac{e^{w_k^\top \phi_n}}{\sum_j e^{w_j^\top \phi_n}}.$$

The likelihood cannot distinguish these weight configurations. At a MAP minimizer, summing the K zero-gradient conditions gives

$$0 = \sum_n \left(\sum_j y_{nj} - \sum_j t_{nj} \right) \phi_n + \alpha R \sum_j w_j = \alpha R \sum_j w_j.$$

Since $\alpha > 0$ and R is invertible,

$$\sum_j w_j = 0.$$

Equivalently, within any common-shift family the quadratic penalty is smallest at $v = -K^{-1} \sum_j w_j$. The prior chooses the centered representative; it does not remove the invariance of the prediction formula itself. Here the positive definite penalty also makes the complete objective strongly convex and coercive, so it has a unique finite MAP minimizer.

Suggested credit: 0.5 p for shift invariance; 0.5 p for the zero-sum proof and its interpretation.

Exercise 2: PCA with an offset and a forbidden direction

2a All affine offsets and the trace objective

[2 p]

Write $P = uu^\top$. Then $P^\top = P$, $P^2 = P$, and

$$x_i - \hat{x}_i = (I - P)(x_i - c).$$

Decompose $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$. The cross term disappears because $\sum_i q_i(x_i - \bar{x}) = 0$, so

$$\begin{aligned} E(c, u) &= \sum_i q_i(x_i - \bar{x})^\top (I - P)(x_i - \bar{x}) + \|(I - P)(\bar{x} - c)\|^2 \\ &= \text{tr}[S(I - P)] + \|(I - P)(\bar{x} - c)\|^2. \end{aligned}$$

For fixed u , all minimizers satisfy $(I - P)(\bar{x} - c) = 0$. Therefore

$$\boxed{c = \bar{x} + \gamma u, \quad \gamma \in \mathbb{R}}.$$

These offsets describe the same affine line. In particular, $c = \bar{x}$ is a valid representative but is not the only one. After minimizing over c ,

$$\boxed{\min_c E(c, u) = \text{tr}[S(I - uu^\top)] = \text{tr}(S) - u^\top Su}.$$

Thus maximizing retained variance minimizes reconstruction error. The repetition weights are required: they represent the full N -observation dataset, not an equally weighted dataset of the J distinct vectors.

Suggested credit: 1 p for the decomposition and all offsets; 1 p for the trace reduction and offset interpretation.

2b Unconstrained component and a new input

[1 p]

An orthonormal eigensystem of S is

$$u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \lambda_1 = 7; \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \lambda_2 = 3; \quad u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \lambda_3 = 1.$$

For a unit vector $u = \sum_j \beta_j u_j$, $u^\top Su = \sum_j \lambda_j \beta_j^2$ and $\sum_j \beta_j^2 = 1$. The largest possible retained variance is 7, attained at $u = \pm u_1$. Choosing $u = u_1$,

$$\boxed{\text{retained variance} = 7, \quad E_{\min} = 11 - 7 = 4}.$$

The new centered input is $r_* = x_* - \bar{x} = (2, 1, 2)^\top$. Consequently

$$z_* = u_1^\top r_* = \frac{3}{\sqrt{2}}, \quad \hat{x}_* = \bar{x} + u_1 z_* = \boxed{\left(\frac{11}{2}, -\frac{1}{2}, 1\right)^\top}.$$

Its residual is $(\frac{1}{2}, -\frac{1}{2}, 2)^\top$, with squared norm $\boxed{9/2}$. A single point's error need not equal the training average 4. Changing the sign of u_1 changes z_* but not the reconstruction.

Suggested credit: 0.5 p for the direction, variance and average error; 0.5 p for the coordinate, reconstruction and pointwise error.

2c The forbidden direction changes the optimum

[2 p]

Here $v = u_1$. Maximize $u^\top Su$ subject to both constraints with

$$\mathcal{L}(u, \lambda, \eta) = u^\top Su - \lambda(u^\top u - 1) - 2\eta v^\top u.$$

The stationary conditions are

$$Su - \lambda u - \eta v = 0, \quad u^\top u = 1, \quad v^\top u = 0.$$

Multiplying the first equation by $v^\top$ and using $Sv = 7v$ gives $\eta = 0$. The feasible plane is spanned by u_2, u_3 . Their eigenvalues are distinct, so the stationary directions and multipliers are exactly

$$u = \pm u_2, \lambda = 3, \eta = 0; \quad u = \pm u_3, \lambda = 1, \eta = 0.$$

Signs or multiplier values can change if a different Lagrangian sign convention is used; the directions and their classification cannot. On the feasible unit circle, writing $u = \beta_2 u_2 + \beta_3 u_3$ gives variance $3\beta_2^2 + \beta_3^2 \leq 3$. Thus $\pm u_2$ are global maxima, while $\pm u_3$ are global minima of retained variance.

$$E_{\min, \text{constrained}} = 11 - 3 = 8.$$

Using u_2 for the new input gives $z_* = 1/\sqrt{2}$ and

$$\hat{x}_* = \bar{x} + \left(\frac{1}{2}, -\frac{1}{2}, 0\right)^\top = \left(\frac{9}{2}, -\frac{5}{2}, 1\right)^\top.$$

The residual is $(\frac{3}{2}, \frac{3}{2}, 2)^\top$, so its squared norm is $17/2$.

Suggested credit: 1 p for the constrained stationary calculation and all directions; 0.5 p for the global maximum and average error; 0.5 p for the new-point calculation.

2d PPCA likelihood and residual noise

[2 p]

In the eigenbasis (u_1, u_2, u_3) , C has eigenvalues $a + \tau, \tau, \tau$. For fixed $\mu = \bar{x}$, the log-likelihood is

$$\begin{aligned} \ell(a, \tau) &= -\frac{3N}{2} \log(2\pi) - \frac{N}{2} \log |C| - \frac{N}{2} \text{tr}(C^{-1}S) \\ &= \text{const} - \frac{N}{2} \left[\log(a + \tau) + 2 \log \tau + \frac{7}{a + \tau} + \frac{4}{\tau} \right]. \end{aligned}$$

To avoid confusing the noise variance with its standard deviation, τ itself is the variance here. Introduce $\nu = a + \tau$, so $\nu \geq \tau > 0$. The derivatives with respect to the independent coordinates are

$$\frac{\partial \ell}{\partial \nu} = -\frac{N}{2} \left(\frac{1}{\nu} - \frac{7}{\nu^2} \right), \quad \frac{\partial \ell}{\partial \tau} \Big|_\nu = -\frac{N}{2} \left(\frac{2}{\tau} - \frac{4}{\tau^2} \right).$$

They vanish at $\nu = 7$ and $\tau = 2$, which satisfy the domain constraint. Each bracketed scalar objective decreases up to this point and increases after it; these are global minima of the negative log-likelihood terms. Hence

$$a = 5, \quad \tau = 2, \quad w = \sqrt{5} u_1.$$

The fitted covariance is

$$C = 5u_1 u_1^\top + 2I_3 = \begin{pmatrix} 9/2 & 5/2 & 0 \\ 5/2 & 9/2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Its eigenvalues are $(7, 2, 2)$, not the sample eigenvalues $(7, 3, 1)$. The leading variance is fitted exactly. In the two discarded directions, isotropic noise must use a common variance, their average $(3 + 1)/2 = 2$. This model therefore need not reproduce the full sample covariance.

Suggested credit: 0.5 p for the likelihood; 1 p for the optimizing calculation and domain check; 0.5 p for C and the discrepancy.

2e A different latent coordinate is not a richer marginal family**[1 p]**

Take the positive choice of w from 2d and set

$$w' = \frac{1}{3}w = \frac{\sqrt{5}}{3}u_1, \quad \mu' = \bar{x} - \frac{2}{3}w.$$

Since $\mathbb{E}[z'] = 2$ and $\text{Var}(z') = 9$, the new marginal has

$$\mathbb{E}[x] = 2w' + \mu' = \bar{x}, \quad \text{Cov}(x) = 9w'w'^\top + 2I_3 = ww^\top + 2I_3 = C.$$

Both constructions are Gaussian, so equal means and covariances establish equal marginal distributions. The negative loading choice is also valid if μ' is adjusted to keep $\mu' + 2w' = \bar{x}$.

Changing a scalar Gaussian latent prior cannot make the model match S : its signal covariance remains rank at most one, and two marginal eigenvalues must equal the isotropic noise variance. All three eigenvalues of S are distinct. More specifically, with the fixed variance 2, $S - 2I$ has eigenvalues $(5, 1, -1)$, so it cannot be a positive semidefinite signal covariance of any Gaussian loading. The reparameterization changes latent coordinates, not the marginal model's expressive power.

Suggested credit: 0.5 p for parameters and preserved marginal; 0.5 p for the rank/eigenvalue argument.

Scope and source alignment

The format follows `exams/homework-based-exam-1.pdf`: two extended exercises, two hours, 18 points, derivations, and follow-up insights. The actual second midterm's question count and topic selection are unknown.

- Exercise 1 develops HW 2 Exercises 1 and 3, practice sets 3–4, and lectures 5–7. New elements are a shared rate across two different sensor distributions, a sensor missing at prediction time, marginal dependence, a full precision matrix, and the softmax common-shift ambiguity.
- Exercise 2 develops HW 2 Exercises 4–5, practice set 5, and lecture 10. New elements are repetition weights, all affine offsets, a forbidden direction, new numerical data, fitting the PPCA likelihood in eigenvalue coordinates, and the difference between reparameterization and increased model capacity.

The foundational operations necessarily recur, but no original homework or practice-set problem, dataset, or complete derivation sequence is reproduced. The questions are intentionally harder than the first midterm; the point total does not imply equal workload.